

Global and Local Projection-Based Reduced-Order Models

2D Burgers and Hypersonic Vehicle Campaigns

Sebastian Ares de Parga³, Radek Tezaur¹, Charbel Farhat^{1,2}

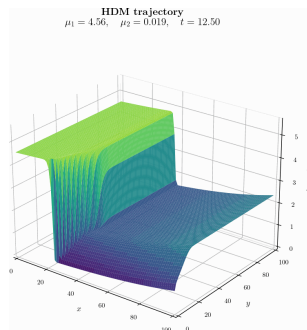
¹Department of Aeronautics and Astronautics

²Institute for Computational and Mathematical Engineering
Stanford University, Stanford, CA 94305, USA

³CIMNE, Barcelona 08034, Spain

September 9, 2026

2D Burgers Campaign



Global and local HPRM validation

Linear POD bases, quadratic manifolds, and GPR manifolds

Setup for Paper Campaign

Two-dimensional parametric inviscid Burgers problem

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x; \boldsymbol{\mu}), \quad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}.$$

$$\mathbf{S}(x; \boldsymbol{\mu}) = \begin{bmatrix} 0.02 \exp(\mu_2 x) \\ 0 \end{bmatrix}, \quad u_x(0, y, t; \boldsymbol{\mu}) = \mu_1.$$

 μ_1 : inflow value μ_2 : exponential source variation $\boldsymbol{\mu} \in [4.25, 5.50] \times [0.015, 0.030]$

Test points:

$$\boldsymbol{\mu}_1 = (4.56, 0.019), \quad \boldsymbol{\mu}_2 = (4.75, 0.020), \quad \boldsymbol{\mu}_3 = (5.19, 0.026).$$

HDM: $\Omega = [0, 100]^2$, 250×250
finite-volume cells.

Methods included:

Global HPROM

Local HPROM

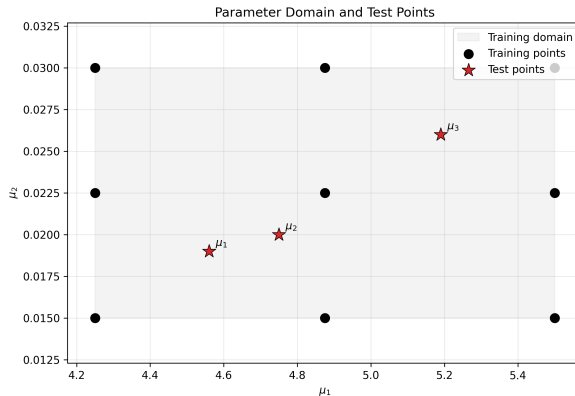
Global HQPROM

Local HQPROM

Global HPROM-GPR

Local HPROM-GPR

Parameter Domain and Test Points



Online Performance (250×250): Avg/Max Error over 3 Test Points

Computational model	n (N for HDM)	$\bar{n}$	N_e	avg $\mathbb{RE}_{2,u}$ (%)	max $\mathbb{RE}_{2,u}$ (%)	Wall-clock time (min)	Speedup
HDM	125 000	–	62 500	–	–	12.517	–
HPROM	96	–	4 824	1.030	1.096	0.905	13.8
HQPROM	39	–	3 505	0.871	0.933	0.967	12.9
HPROM-GPR	20	131	1 801	0.691	0.845	0.407	30.8
Local HPROM ($N_c = 3$)	35–48	–	3 405	0.897	0.933	0.364	34.4
Local HQPROM ($N_c = 3$)	21–27	–	2 152	0.888	1.025	0.504	24.8
Local HPROM-GPR ($N_c = 3$)	9	60–97	811	0.714	0.786	0.284	44.1
Local HPROM ($N_c = 5$)	23–31	–	2 384	0.915	0.943	0.265	47.3
Local HQPROM ($N_c = 5$)	15–20	–	1 593	0.760	0.878	0.380	33.0
Local HPROM-GPR ($N_c = 5$)	7	41–68	631	0.673	0.880	0.251	49.8
Local HPROM ($N_c = 8$)	16–26	–	1 781	0.787	0.900	0.220	56.9
Local HQPROM ($N_c = 8$)	10–15	–	1 075	0.787	0.890	0.296	42.3
Local HPROM-GPR ($N_c = 8$)	6	28–54	541	0.801	1.156	0.243	51.4

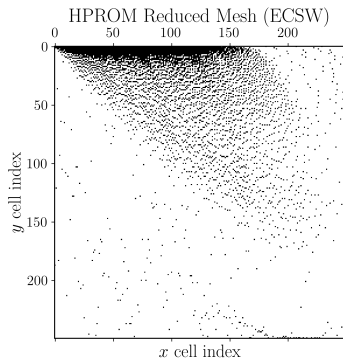
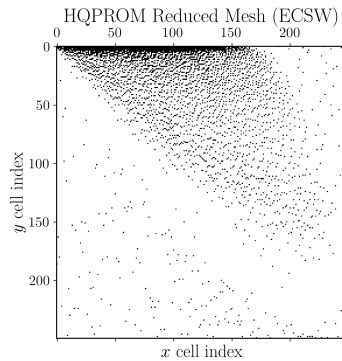
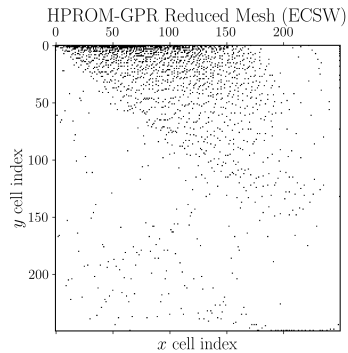
Speedup is computed with respect to the HDM runtime, $t_{\text{HDM}} = 751.043$ s, averaged over the three test points.

Online timings were measured on Sherlock using one node, 24 MPI ranks, and 24 cores.

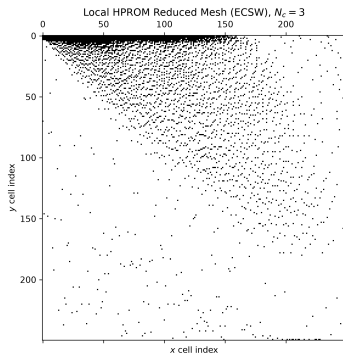
N_e denotes the number of sampled ECSW elements ($N_e = 62,500$ for HDM/full mesh).

All models are reported at a matched accuracy target of $1\% \pm 0.2\%$ in $\max \mathbb{RE}_{2,u}$: for each N_c the accuracy knob is tuned into that window (pod.tol for Local HPROM, ζ_{qua} for Local HQPROM, n for Local HPROM-GPR), so differences in n , N_e and speedup are read at constant accuracy.

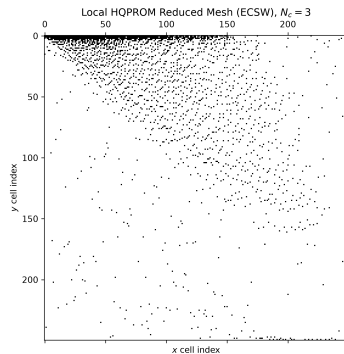
Global Reduced Meshes (ECSW)

HPROM ($N_e = 4,824$)HQPROM ($N_e = 3,505$)HPROM-GPR ($N_e = 1,801$)

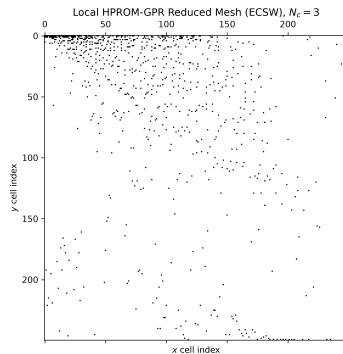
Local Reduced Meshes (ECSW), $N_c = 3$



Local HPROM ($N_e = 3,405$)



Local HQPROM ($N_e = 2,152$)

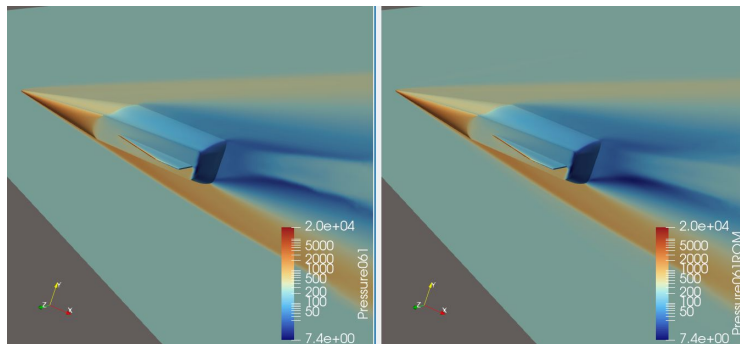


Local HPROM-GPR ($N_e = 811$)

Burgers Conclusions

- ▶ All local ROMs improve runtime performance relative to their global counterparts, at every N_c and at matched accuracy: $13.8\times \rightarrow 56.9\times$ (HPROM), $12.9\times \rightarrow 42.3\times$ (HQPPROM), $30.8\times \rightarrow 51.4\times$ (HPROM-GPR).
- ▶ Increasing N_c monotonically improves speedup within every family, so clustering, not the choice of local manifold, is the dominant lever.
- ▶ Local HPROM at $N_c = 8$ is the fastest model: 0.220 min and $56.9\times$ speedup.
- ▶ Compactness and wall-clock time rank differently. At $N_c = 8$, HPROM-GPR is the most compact ($n = 6$, $N_e = 541$) and HQPPROM next ($n = 10-15$, $N_e = 1,075$), yet plain HPROM is fastest despite $n = 16-26$: the per-iteration cost of the quadratic term and its Jacobian, and of the GPR evaluation, offsets their dimensional advantage.
- ▶ Reported local-model runtimes include cluster selection and any cluster-transfer cost. The decision and transfer rules are formulation-specific and are defined in the appendix.

Hypersonic Vehicle Campaign

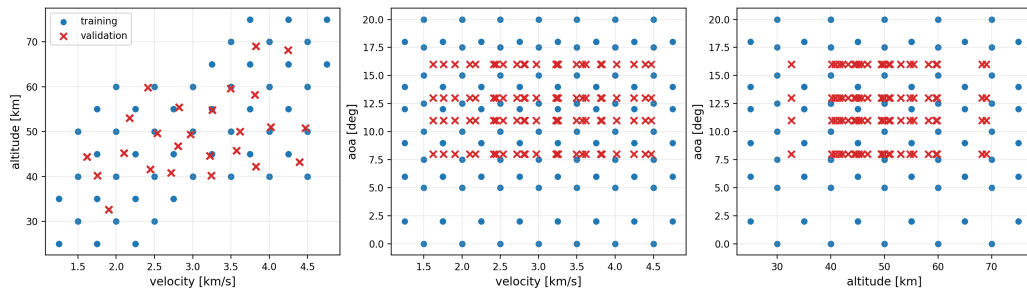


Global and local HPRM validation

Linear POD bases and RBF manifolds

Training and Validation Space

363 HDM training cases; 100 independent HDM validation cases; four local clusters with 20% overlap; ECSW tolerance 10^{-3} .



HGV2 ROM Configurations

- ▶ Global linear HPROM uses one POD basis. Local linear HPROM uses four POD bases and assigns each online state to its nearest cluster.
- ▶ Global HPROM–RBF uses $n = 8$ primary coordinates and predicts the remaining coordinates with one RBF map.
- ▶ Local HPROM–RBF uses one $n = 8$ RBF map per cluster. The local primary and secondary dimensions are reported in the next table.
- ▶ Every POD basis and RBF map is trained from the same 363 HDM parameter cases. ECSW is an independent offline construction step.

HGV2 Online Performance and Validation Accuracy

Coefficient errors are global relative L^2 values over 100 independent validation cases. Online times are means over 10 validation cases on Sherlock: one node, 24 MPI ranks, and 24 cores.

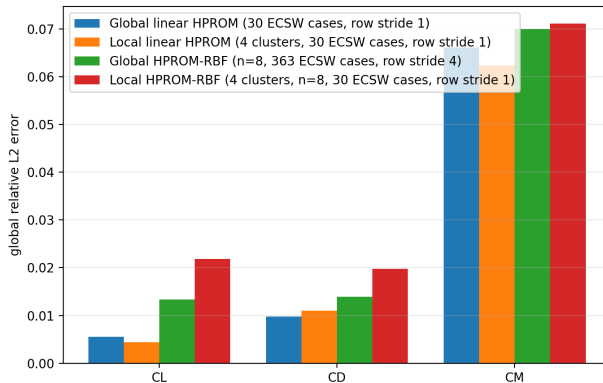
Model	n	$\bar{n}$	C_D [%]	C_L [%]	C_M [%]	Online time [s]		Speedup	
						True	False	True	False
Global linear HPROM	362	–	0.975	0.554	6.620	24.03	–	1.00	–
Local linear HPROM, $N_c = 4$	95–168	–	1.101	0.444	6.242	6.80	–	3.53	–
Global HPROM–RBF	8	354	1.388	1.335	7.004	6.00	1.15	4.00	20.97
Local HPROM–RBF, $N_c = 4$	8	117–140	1.978	2.183	7.117	2.28	0.62	10.54	38.88

True/False denotes IncludeAllLiftDragFaces. For linear HPROM, n is the full POD rank. For RBF, n and $\bar{n}$ are the primary and secondary dimensions; local $\bar{n}$ values are ranges across the four cluster bases. Each speedup is relative to global linear HPROM with True.

The RBF False runs omit the postprocessing-only lift/drag faces and their C_L , C_D , C_M output. The accuracy entries come from the matching True runs. All timing values are means of the maximum-rank *Nonlinear ROM Online* timer over validation cases 000–009.

HGV2 Validation Error Comparison

The plotted global RBF uses all 363 ECSW parameter cases with row stride 4.



HGV2 ECSW Offline Cost

ECSW was run on Sherlock using 20 nodes, 480 MPI ranks, and 480 cores. Times are maximum-rank wall times from the offline logs.

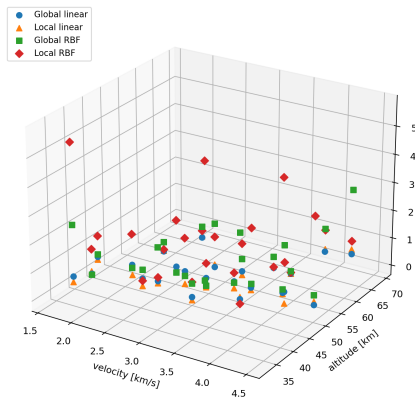
Model	Parameter cases	Rows per case	Row stride	Rows read	Sample nodes	ECSW [h]
Global linear HPROM	30	30^2	1	27,000	5,089	30.04
Local linear HPROM, $N_c = 4$	30	30^2	1	27,000	3,484	9.95
Global HPROM-RBF, $n = 8$	363	8^2	4	5,808	1,715	0.43
Local HPROM-RBF, $N_c = 4$, $n = 8$	30	8^2	1	1,920	1,113	0.38

What the ECSW columns mean

Parameter cases = 30 means a selected subset of the 363 input configurations. For the linear models, $m_k = \min(r_k, 30) = 30$, hence 30^2 candidate Jacobian rows per selected case. For RBF, $m_k = n = 8$, hence 8^2 rows. Row stride 4 reads rows 0, 4, 8, ... within every one of the 363 global-RBF cases: $363 \times 8^2/4 = 5,808$ rows. It does not subsample parameter cases or online Newton iterations.

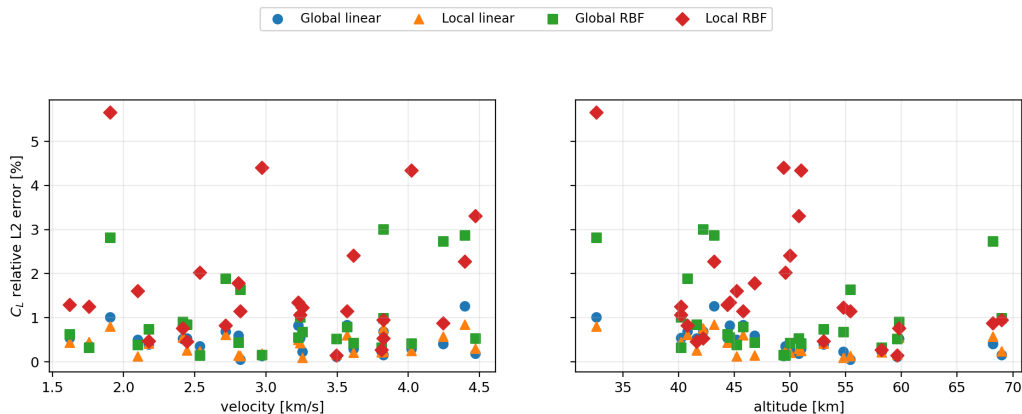
HGV2 Three-Dimensional C_L Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_L error over the four sampled angles of attack.



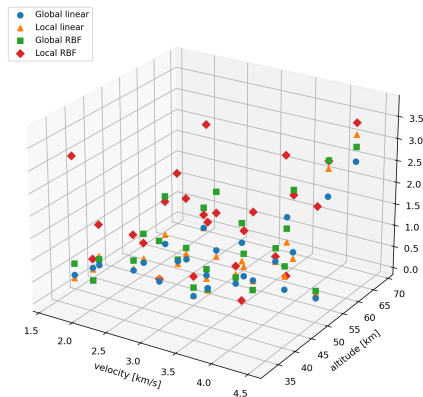
HGV2 Two-Dimensional C_L Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



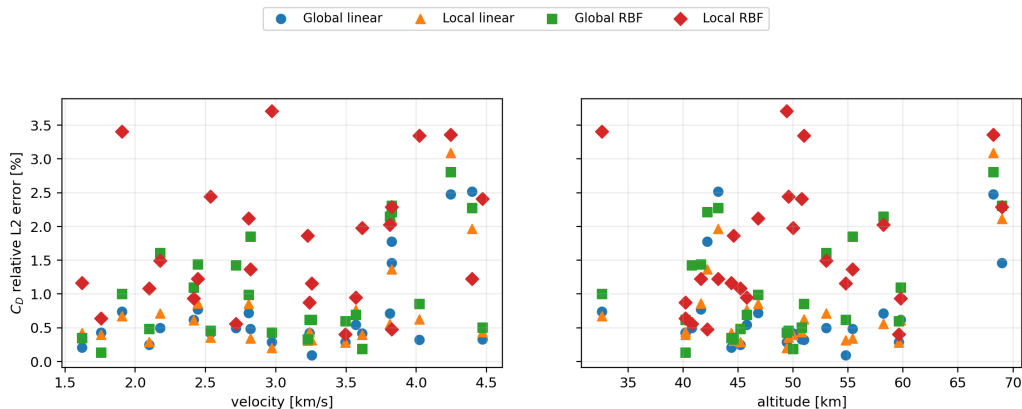
HGV2 Three-Dimensional C_D Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_D error over the four sampled angles of attack.



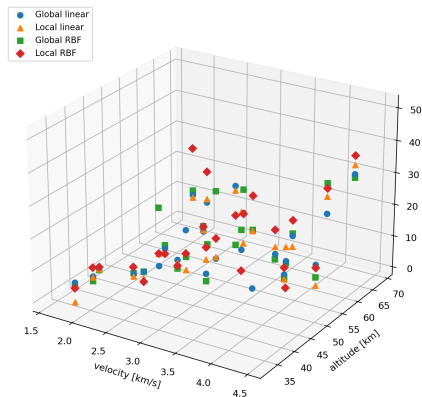
HGV2 Two-Dimensional C_D Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



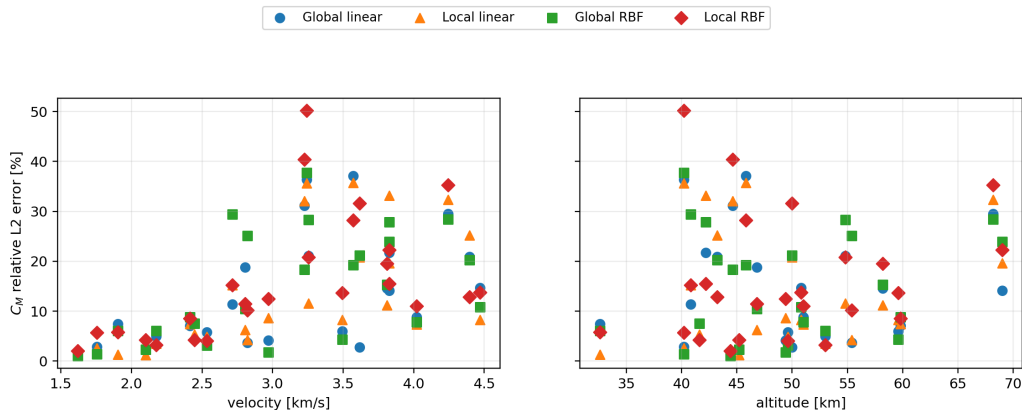
HGV2 Three-Dimensional C_M Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_M error over the four sampled angles of attack.



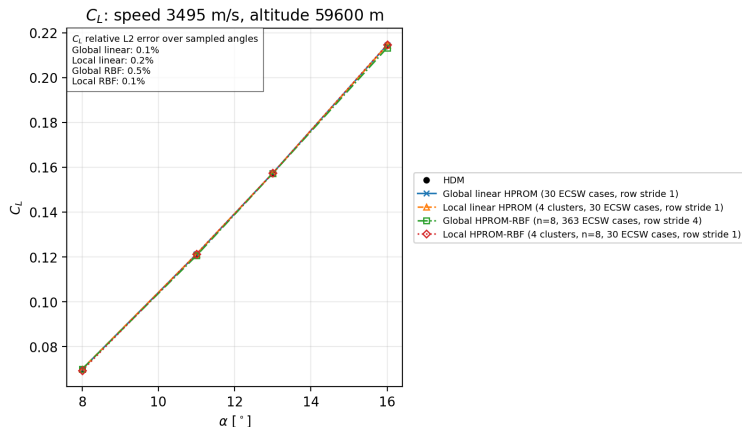
HGV2 Two-Dimensional C_M Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



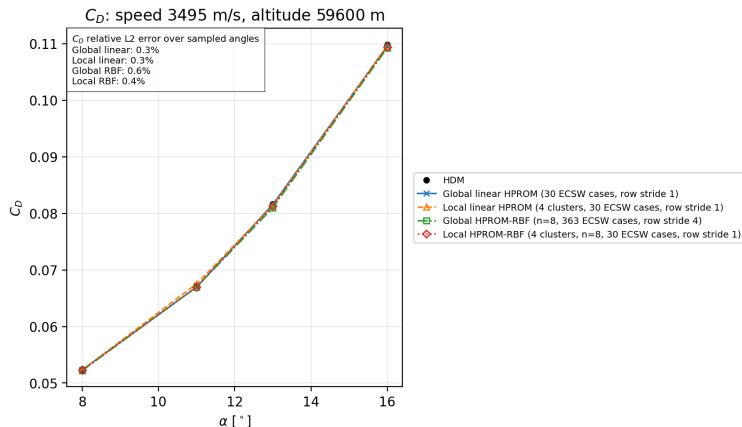
HGV2 Representative Median Pair: C_L

$v = 3.495$ km/s, altitude = 59.6 km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L , C_D , and C_M relative-error norm.



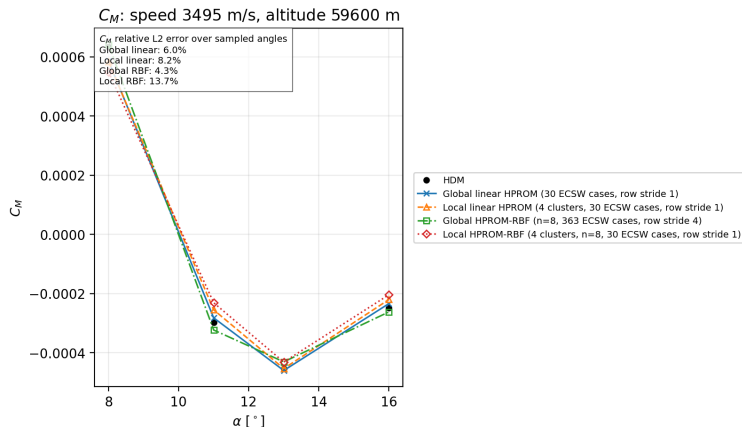
HGV2 Representative Median Pair: C_D

$v = 3.495$ km/s, altitude = 59.6 km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L , C_D , and C_M relative-error norm.



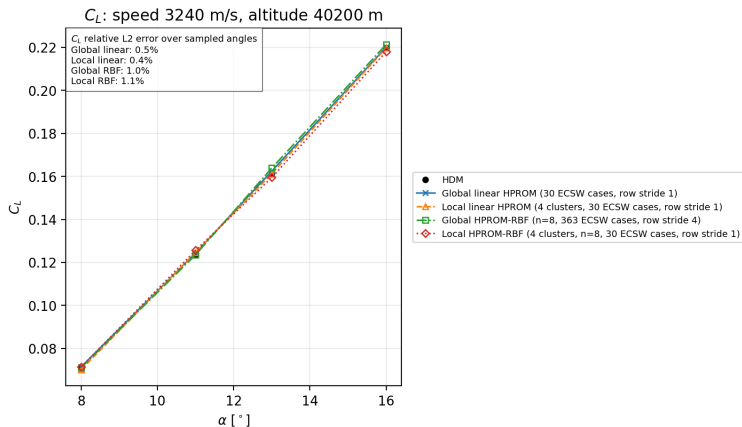
HGV2 Representative Median Pair: C_M

$v = 3.495$ km/s, altitude = 59.6 km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L , C_D , and C_M relative-error norm.



HGV2 Representative Challenging Pair: C_L

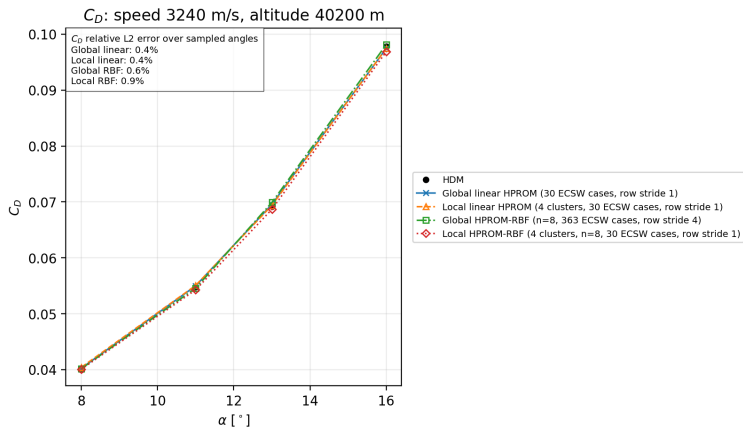
$v = 3.240$ km/s, altitude = 40.2 km. This pair has the largest combined local-RBF error across



C_L , C_D , and C_M .

HGV2 Representative Challenging Pair: C_D

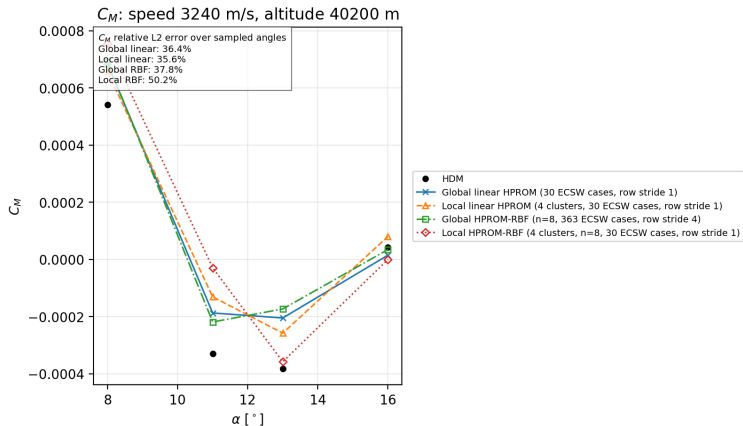
$v = 3.240$ km/s, altitude = 40.2 km. This pair has the largest combined local-RBF error across



C_L , C_D , and C_M .

HGV2 Representative Challenging Pair: C_M

$v = 3.240$ km/s, altitude = 40.2 km. This pair has the largest combined local-RBF error across



C_L , C_D , and C_M .

HGV2 Conclusions

- ▶ Global and local linear HPROM give the best validation accuracy. Global linear is the best in C_D ; local linear is the best in C_L and C_M .
- ▶ The global RBF with 363 ECSW cases and row stride 4 is stable and reduces the global-RBF moment error relative to the 30-case, stride-1 ablation, but it does not surpass the linear models.
- ▶ The local RBF is the fastest online model. Its C_M error is comparable to global RBF, but its C_L and C_D errors are larger. The next controlled test should vary one ECSW factor at a time and broaden the RBF kernel/hyperparameter search.

Supplementary results and implementation details

Appendix: ECSW Training Data

For `DataType = Jacobian`, each selected parameter solution assigned to cluster k supplies m_k^2 candidate ECSW rows, where $m_k = \min(r_k, \text{MaxNumResidualComponents})$ and `MaxNumResidualComponents` = 30. With row stride s , CM reads every s -th row of its `training_rows` sequence.

Model	ECSW cases	Row stride	m_k	Rows read by CM
Global linear HPROM	30	1	$\min(362, 30) = 30$	$30 \times 30^2 = 27,000$
Local linear HPROM, $N_c = 4$	30	1	$\min(r_k, 30) = 30$	$30 \times 30^2 = 27,000$
Global HPROM-RBF, $n = 8$ (ablation)	30	1	$\min(8, 30) = 8$	$30 \times 8^2 = 1,920$
Global HPROM-RBF, $n = 8$ (final)	363	4	$\min(8, 30) = 8$	$363 \times 8^2/4 = 5,808$
Local HPROM-RBF, $N_c = 4$, $n = 8$	30	1	$\min(8, 30) = 8$	$30 \times 8^2 = 1,920$

Interpretation

An ECSW snapshot is a Jacobian-training row, not a physical-time state. The RBF primary dimension $n = 8$ —not $\bar{n}$ —sets the 8×8 count. A row stride of 4 reads rows 0, 4, 8, . . . within every selected parameter case; it does not subsample the 363 parameter cases and does not refer to online Newton iterations.

Appendix: HGV2 Reduced-Mesh Construction

Model	L0	L1	L2	L3	Nodes	Elms.	N_{-L2}
Global linear HPROM	5,089	36,971	166,878	60,623	269,561	563,301	92,683
Local linear HPROM, 4 clusters	3,484	28,145	169,798	47,183	248,610	470,649	78,812
Global HPROM-RBF, $n = 8$	1,715	15,809	173,072	28,648	219,244	350,921	46,172
Local HPROM-RBF, 4 clusters, $n = 8$	1,113	10,516	175,813	20,602	208,044	298,219	32,231

L0: selected ECSW sample nodes.

L1: nodes added by edge connectivity for linear reconstruction.

L2: lift-surface nodes used only for C_L , C_D , and C_M postprocessing.

L3: nodes added by element connectivity for state reconstruction.

N_{-L2} excludes the postprocessing-only lift-surface nodes.

Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual $\mathbf{r}(\mathbf{u}) = \mathbf{0}$, define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^* = \arg \min_{\mathbf{q}} \frac{1}{2} \|\mathbf{r}(\mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}))\|_2^2.$$

With Gauss-Newton (current iterate $\mathbf{q}^{(k)}$), the increment $\Delta \mathbf{q}$ is obtained from

$$(\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{J} \boldsymbol{\Psi}) \Delta \mathbf{q} = -\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{r},$$

where

$$\boldsymbol{\Psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}}, \quad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \quad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta \mathbf{q}.$$

Global Linear HPRM

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}, \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \quad n = 96.$$

Hence

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \quad \boldsymbol{\Psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}) \Delta \mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Global HQPROM

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}), \quad n = 39.$$

Thus

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}),$$

and the tangent depends on the current reduced state:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}[\mathbf{H}(\mathbf{q} \otimes \mathbf{q})].$$

Global HPROM-GPR

Split basis and reduced coordinates:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{q}_{\text{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\text{GPR}}(\mathbf{q}), \quad n = 20, \bar{n} = 131.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}),$$

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\text{tot}} \mathbf{q}_{\text{tot}} = \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}_{\text{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}_{\text{GPR}}}{\partial \mathbf{q}}.$$

Local ROMs: Piecewise Manifolds

Partition the training snapshots into clusters $k = 1, \dots, N_c$ and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q},$$

$$\Phi_k^{\text{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\text{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \bar{\mathbf{V}}_k \mathcal{N}_{\text{GPR},k}(\mathbf{q}).$$

Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

Cluster Selection: Final Form (Linear)

Shared nearest-centroid selector (used in all local models):

$$k^+ = \arg \min_{\ell=1,\dots,N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2, \quad \mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

For the linear local manifold $\Phi_{k^-}^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}$:

$$k^+ = \arg \min_{\ell=1,\dots,N_c} S_{k^-, \ell}^{\text{lin}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{lin}}(\mathbf{q}) = 2\mathbf{g}_{k^-, \ell}^T \mathbf{q} + d_{k^-, \ell},$$

with offline-precomputed terms

$$\mathbf{g}_{k^-, \ell} := \mathbf{V}_{k^-}^T (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \quad d_{k^-, \ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Cluster Selection: Final Form (HQPROM and HPROM-GPR)

HQPROM selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{quad}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{quad}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) + d_{k^-, \ell},$$

$$\mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell})^T \mathbf{H}_{k^-} (\mathbf{q} \otimes \mathbf{q}).$$

HPROM-GPR selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \bar{\mathbf{g}}_{k^-, \ell}^T \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}) + d_{k^-, \ell}.$$

$$\bar{\mathbf{q}}_{k^-} := \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}_{k^-}), \quad \bar{\mathbf{g}}_{k^-, \ell} := \bar{\mathbf{V}}_{k^-}^T (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell}).$$

Cluster Change: Linear and Nonlinear Transfers

$k^+ = k^- \Rightarrow$ no transfer, $k^+ \neq k^- \Rightarrow$ solve transfer problem.

Transferred state:

$$\mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

Linear local target

$$\mathbf{q}_{k^+}^{\text{lin}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - (\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+} \mathbf{q}) \right\|_2^2,$$

$$\text{if } \mathbf{V}_{k^+}^T \mathbf{V}_{k^+} = \mathbf{I}, \quad \mathbf{q}_{k^+}^{\text{lin}} = \mathbf{V}_{k^+}^T (\mathbf{u}^s - \mathbf{u}_{0,k^+}).$$

Nonlinear local targets (HQPROM / HPROM-GPR)

$$\mathbf{q}_{k^+}^{\text{nl}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - \Phi_{k^+}^{\text{nl}}(\mathbf{q}) \right\|_2^2.$$

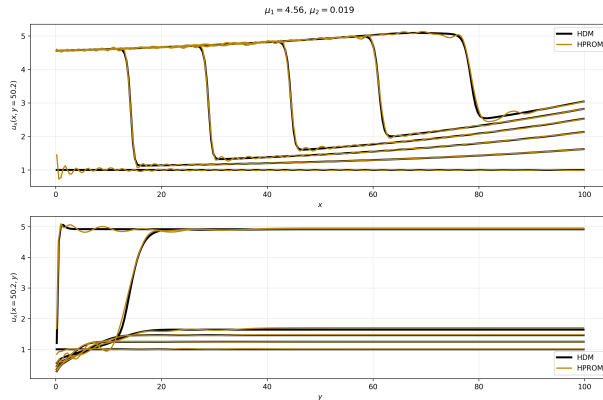
Cluster Change: Nonlinear Least Squares

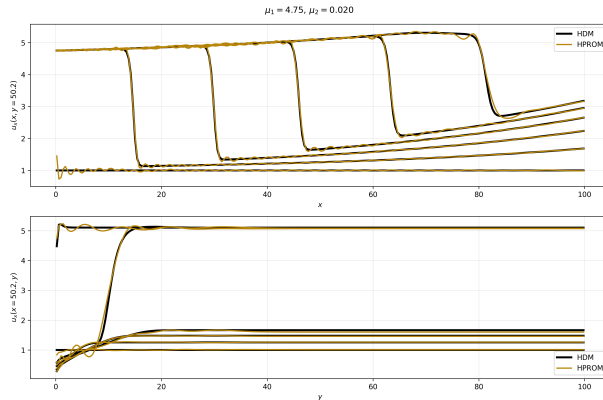
For nonlinear local manifolds, define

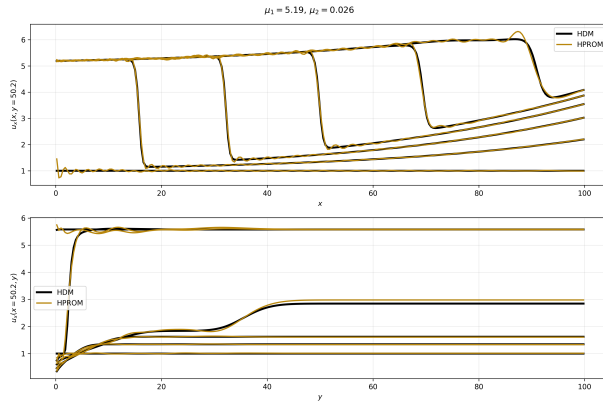
$$\mathbf{r}_{\text{tr}}(\mathbf{q}) := \Phi_{k^+}(\mathbf{q}) - \mathbf{u}^s, \quad \mathbf{J}_{\Phi}(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}}.$$

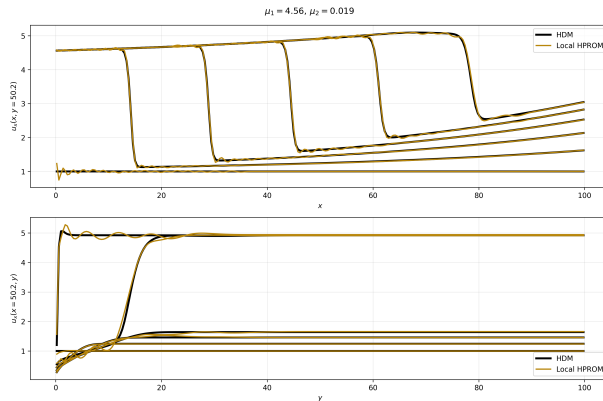
Gauss–Newton at iterate $\mathbf{q}^{(j)}$:

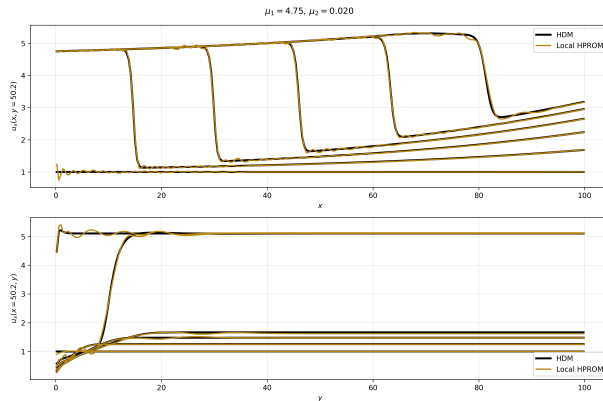
$$(\mathbf{J}_{\Phi}^T \mathbf{J}_{\Phi}) \Delta \mathbf{q} = -\mathbf{J}_{\Phi}^T \mathbf{r}_{\text{tr}}(\mathbf{q}^{(j)}), \quad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta \mathbf{q}.$$

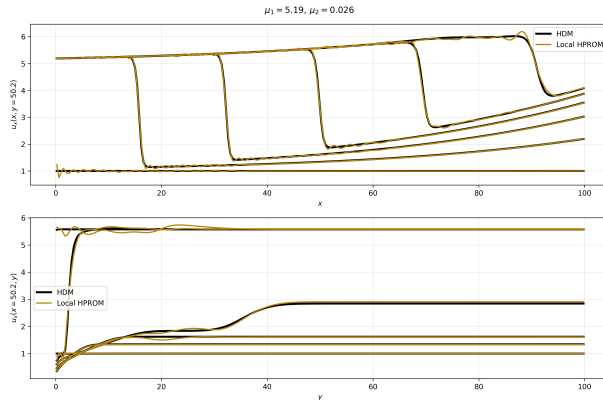
HPROM vs HDM: $\mu = (4.56, 0.019)$ 

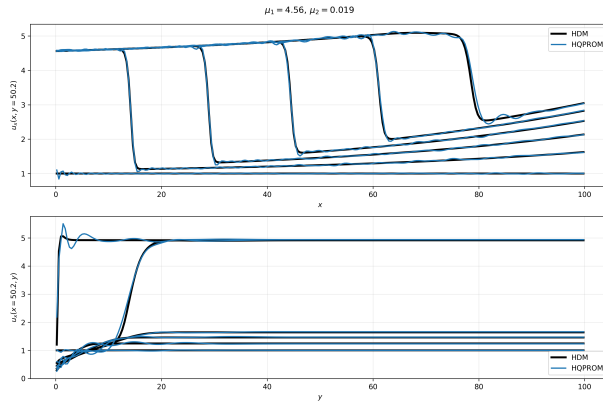
HPROM vs HDM: $\mu = (4.75, 0.020)$ 

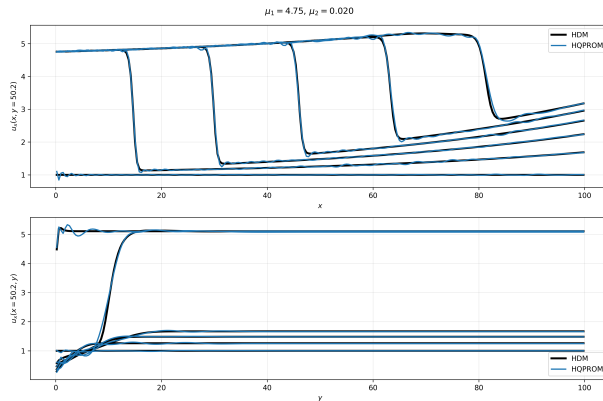
HPROM vs HDM: $\mu = (5.19, 0.026)$ 

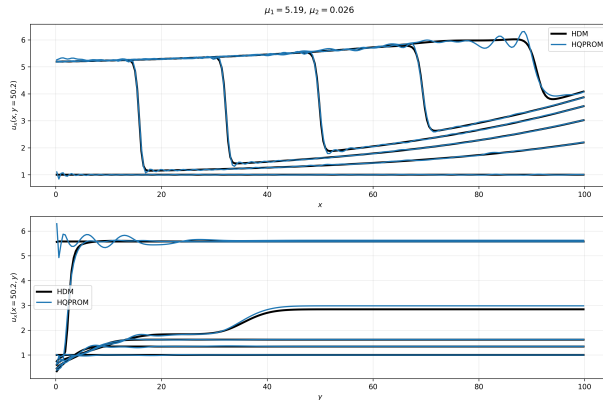
Local HPROM ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$ 

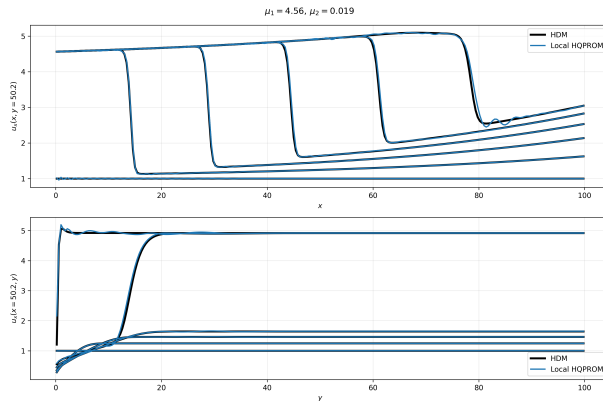
Local HPROM ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$ 

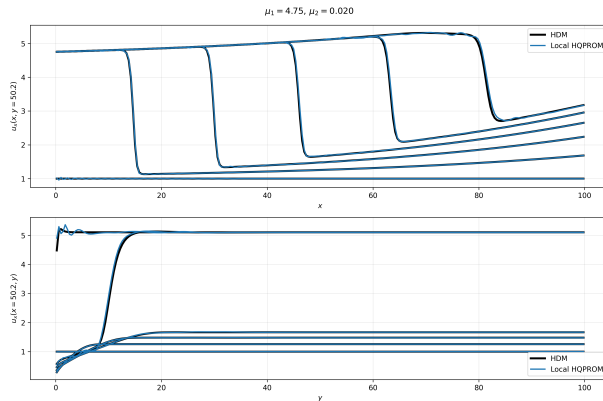
Local HPROM ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$ 

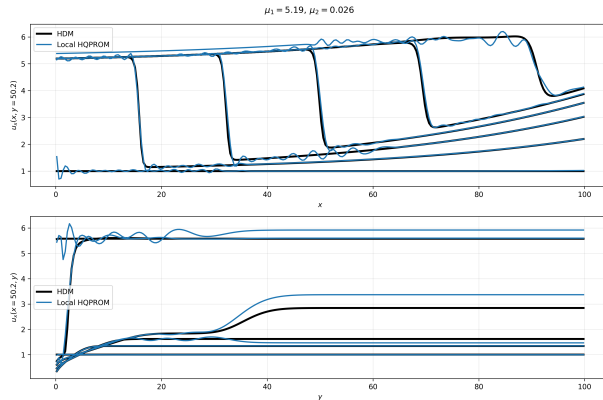
HQPROM vs HDM: $\mu = (4.56, 0.019)$ 

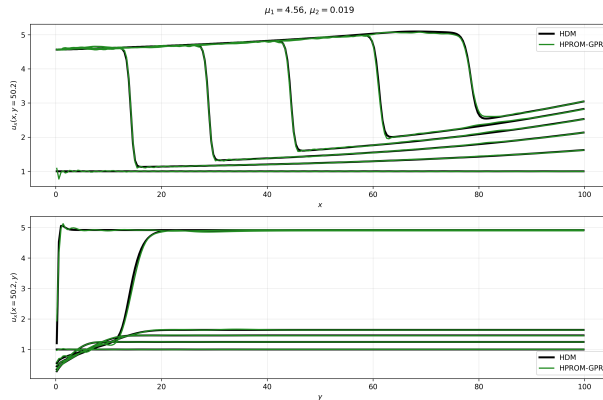
HQPROM vs HDM: $\mu = (4.75, 0.020)$ 

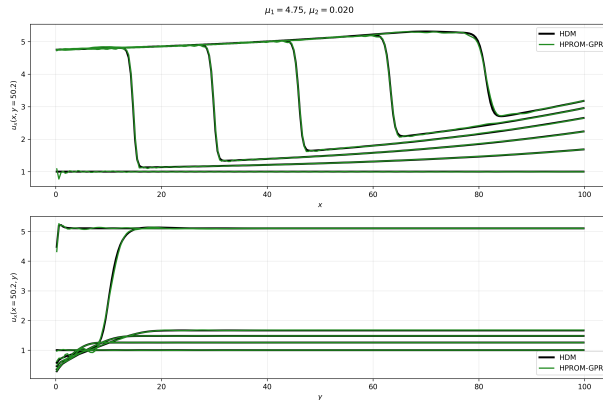
HQPROM vs HDM: $\mu = (5.19, 0.026)$ 

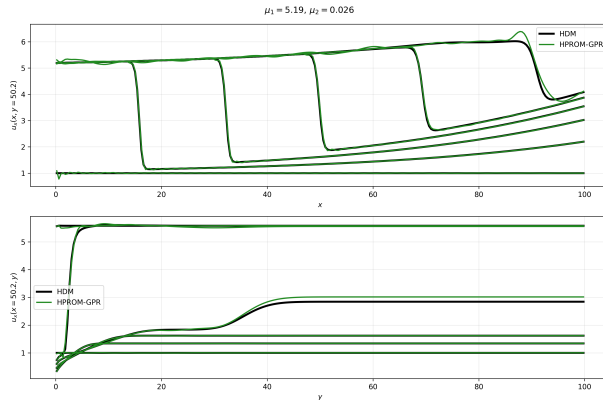
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$ 

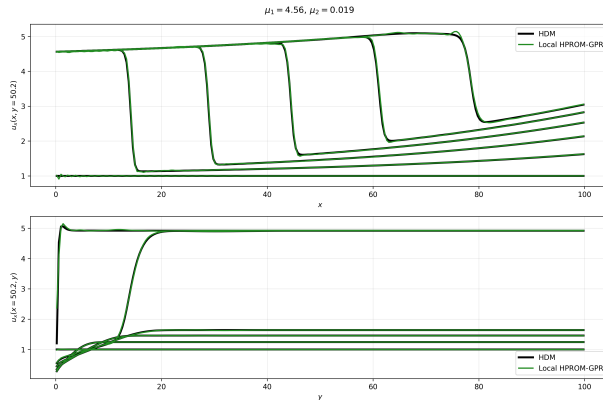
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$ 

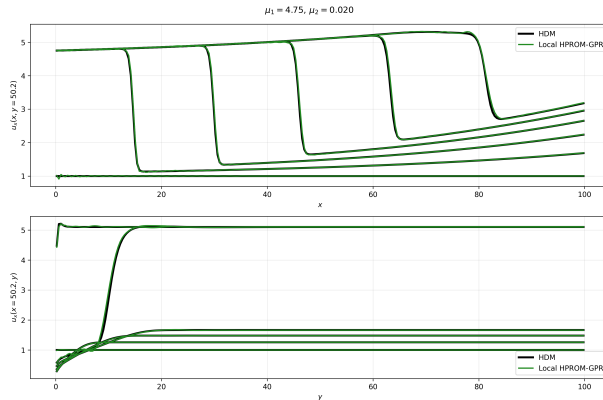
Local HQPROM ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$ 

HPROM-GPR vs HDM: $\mu = (4.56, 0.019)$ 

HPROM-GPR vs HDM: $\mu = (4.75, 0.020)$ 

HPROM-GPR vs HDM: $\mu = (5.19, 0.026)$ 

Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (4.56, 0.019)$ 

Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (4.75, 0.020)$ 

Local HPROM-GPR ($N_c = 3$) vs HDM: $\mu = (5.19, 0.026)$ 